SDS 192 Midterm Exam

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Instructions:

- This midterm lasts 75 minutes.
- This exam is closed-book, to be individually completed and without the aid of the internet or mobile phones. You may reference the 'R' cheatsheets, along with any supplementary material I have provided.
- If there are any areas of the exam that are ambiguous to you, please note that on the exam, state your assumptions, and use your best judgment.

Several questions in this exam will reference an example dataset. That dataset is the 2018 Central Park Squirrel Census. I have provided you with the documentation for that dataset. Please note that you will be working specifically with the **Hectare Data**.

1. What is the unit of observation in the example dataset?

2. Identify one nominal, one ordinal, one discrete numeric variable in the example dataset.

3. What are the characteristics of a data frame in `R`? What are the characteristics of a vector? How are they related?

4. What is the chief difference between a barplot and a histogram?

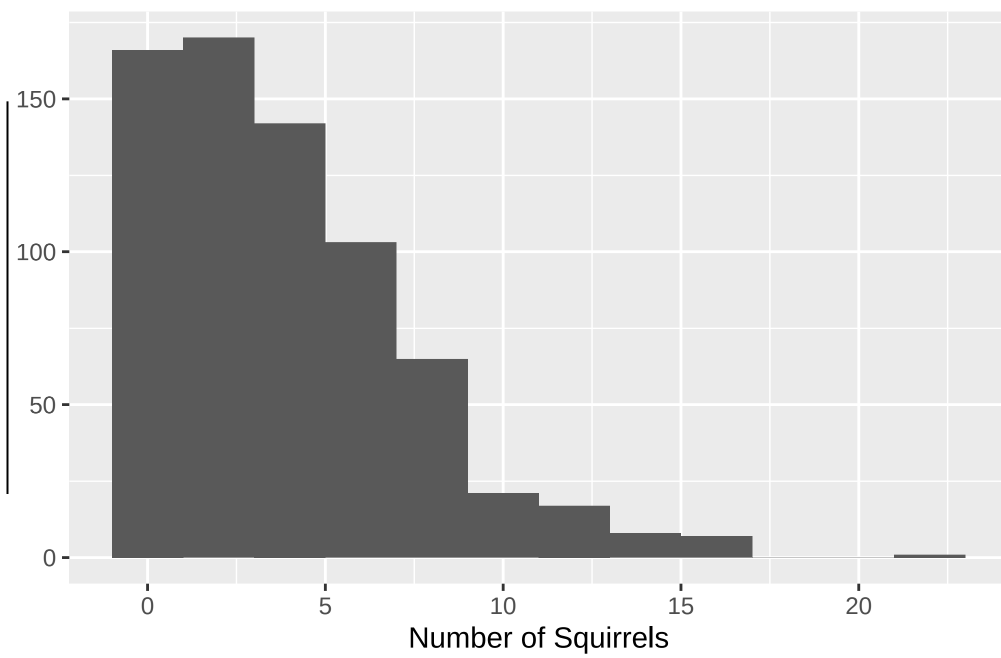
5. We saw in class that there are two ways to create a barplot using `ggplot`: either using `geom_bar()` or using `geom_col()`. Illustrate the need for these two different ways using simple data frame examples.

6. What is overplotting? What are two ways that we can deal with overplotting?

7. When would we use each of the following types of color palettes on a plot: a qualitative palette, a sequential palette, and a divergent palette?

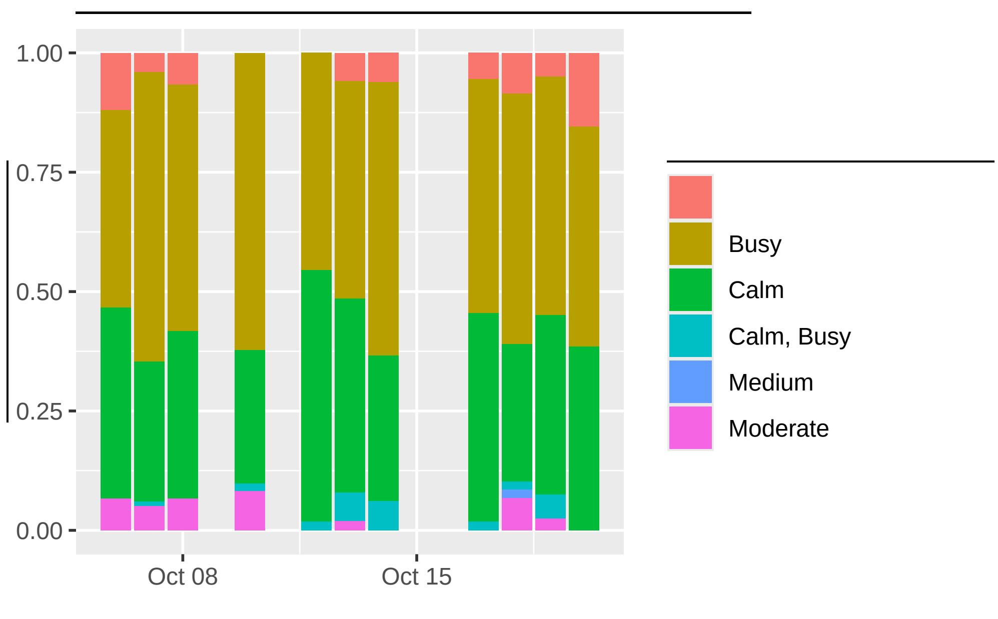
8. Given the example dataset, what would be an appropriate y-axis label for the following plot?

```
nyc_squirrel_census |>
  ggplot(aes(x = number_of_squirrels)) +
  geom_histogram(binwidth = 2) +
  labs(x = "Number of Squirrels",
       y = "________________________")
```



9. Develop a descriptive title and labels for the following plot.

```
library(lubridate)
nyc_squirrel_census |>
  mutate(date = mdy(date)) |>
  ggplot(aes(x = date, fill = hectare_conditions)) +
  geom_bar(position = "fill") +
  labs(title = "___________________________________",
       x = "________________",
       y = "________________",
       fill = "____________________")
```

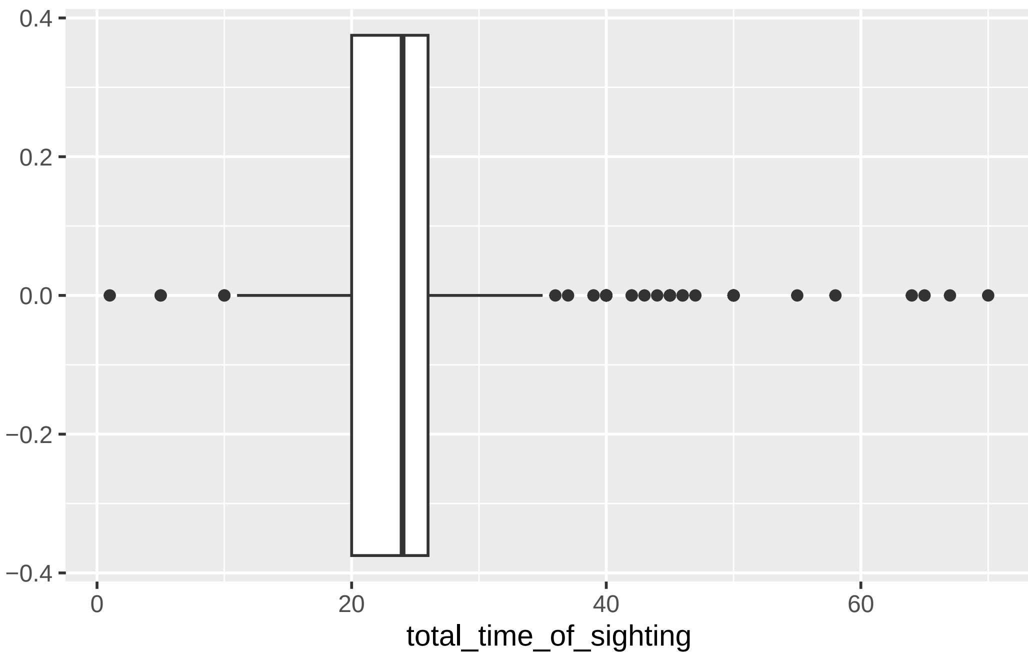


10. What is the scale of the x-axis on the previous plot? What is the scale of the y-axis?

11. According to Tufte, what is a plot's lie factor?

12. Approximately what is the interquartile range represented on the following plot?

```r
nyc_squirrel_census |>
  ggplot(aes(x = total_time_of_sighting)) +
  geom_boxplot()
```



13. Two students were working on the same lab assignment and tried to edit the same line in the file. The merge conflict below appeared. What should Student 2 do to ensure that their version of the code gets pushed to the remote repository?

```
<<<<<<< HEAD:lab1.Rmd
#Student 1's code
ggplot(lobby_chicago, aes(x = ACTION)) + geom_bar()
-------
#Student 2's code
```

```
ggplot(lobby_chicago, aes(x = PURPOSE)) + geom_bar()
>>>>>>> hash
```

14. Fill in the following statement with the correct `git` vocabulary.

`git __________` downloads a remote repository to your local computer. `git __________` fetches changes to code from a remote repository to your local machine. ______________ involves selecting files we wish to commit changes on. `git __________` saves a snapshot of the current version of code to the git version control. `git __________` uploads changes to a remote repository.

15. What is branching? Outline a collaborative GitHub workflow that engages branching to incorporate multiple changes into a final document.

16. In what hectares and on what dates and shifts did anonymous sighter 37 count? Write pseduo-code to produce the following data frame.

```
  hectare shift     date anonymized_sighter
1     11G    AM 10122018                 37
2     11H    AM 10102018                 37
3     14G    AM 10122018                 37
4     16G    AM 10102018                 37
5     18I    AM 10172018                 37
6     21H    AM 10172018                 37
7     21I    AM 10182018                 37
8     22H    AM 10182018                 37
```

```
9       38I     AM 10192018                    37
10      39I     AM 10192018                    37
```

17. What would be the difference in outputs between the following two code chunks?

```r
mean(nyc_squirrel_census$total_time_of_sighting, na.rm = TRUE)

nyc_squirrel_census |>
  summarize(mean_time_of_sighting = mean(total_time_of_sighting, na.rm = TRUE))
```

18. In which 10 counting sessions where the most squirrels sighted per minute? Return for me their hectares, shifts, dates, and sightings per minute. Write pseudo-code to produce the following data frame.

```
   hectare shift     date sightings_per_minute
1      35F    AM 10072018            1.0000000
```

```
2       04C    PM 10102018             0.8500000
3       07F    AM 10082018             0.7727273
4       01B    AM 10122018             0.7391304
5       03B    PM 10072018             0.7333333
6       12F    PM 10072018             0.7142857
7       20F    PM 10072018             0.7000000
8       32E    AM 10072018             0.7000000
9       05C    PM 10102018             0.6500000
10      33E    AM 10142018             0.6500000
```

19. In which 5 hectares were the most total squirrels squirrels sighted? How much time was spent sighting in each of these hectares? Write pseduo-code to produce the following data frame.

```
# A tibble: 5 x 3
  hectare total_squirrels total_time_hectare
  <chr>             <int>              <int>
1 14D                  32                 90
2 32E                  30                 45
3 14E                  28                 65
4 01B                  27                 58
5 07H                  26                 55
```

20. What was the total, mean, median time spent sighting squirrels in AM vs. PM counting sessions, and how many times was the total time spenting counting recorded as `NA`? Write pseduo-code to produce the following data frame.

```
# A tibble: 2 x 5
  shift total_time mean_time median_time na_time
  <chr>      <int>     <dbl>       <int>   <int>
1 AM          8457      24.9          24      11
2 PM          8480      24.9          24       9
```